Venkata Lakshmi Parimala Pasupuleti

Senior Software Engineer · RAG Pipelines · Agentic Workflows · LLM Evaluation · MCP · Vector Search · Full-Stack · Python · TypeScript · AWS · CI/CD · Observability

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PROFESSIONAL SUMMARY

Senior Software Engineer with two production AI systems that cover every dimension of Qualcomm's Enterprise AI Platform stack - and 8 years of production Python engineering at scale. AgentProbe is an open-source LLM evaluation framework with tracing and observability for AI applications: 5-metric quality engine (correctness, groundedness, latency, token efficiency, hallucination rate), LLM-as-judge scoring, CI-integrated regression gate, multi-provider routing - the exact "evaluation frameworks, observability, tracing, and operational tooling" the JD requires. LeaseI is an 8-agent LangGraph agentic platform with full RAG pipeline (document ingestion, OCR, indexing, embeddings, vector search, reranking, retrieval optimization, grounding) and MCP-compatible tool interfaces - 93.6% precision across 1,240 compliance test cases. Entrata background provides 8yr production Python at 100K+ MAU, 99.5% uptime: cloud-native AWS infrastructure, event-driven distributed systems, FastAPI REST APIs, Docker containerization, and CI/CD. Full-stack delivery across Python, TypeScript, JavaScript, and React. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

AI & RAG Engineering: RAG Pipelines (document ingestion, indexing, embeddings, vector search, reranking, retrieval optimization, grounding), Agentic Workflows, LangGraph, LangChain, Multi-Agent Orchestration, MCP (Model Context Protocol), LLM Evaluation Frameworks, LLM-as-Judge, LLM Observability & Tracing, Hallucination Detection, HITL, Guardrails, Structured Outputs, Semantic Search, ChromaDB, TF-IDF, Embeddings, Anthropic API, OpenAI API, Claude Code, BERT, spaCy, NLP

Full-Stack & Backend: Python (8yr), TypeScript, JavaScript, React.js, Node.js, FastAPI, Flask, REST APIs, HTML/CSS, SQL, Microservices, Docker, CI/CD (GitHub Actions), AWS (S3, EC2, RDS, Glue - cloud-native), RabbitMQ, Redis, PostgreSQL, MySQL, Oracle, NoSQL, Bash, Linux, Git

Reliability & Platform Engineering: CI/CD Pipelines, Observability, Production Monitoring, Performance Engineering, Distributed Systems, Data Pipelines, ETL, Batch Processing, PHI Compliance (Libsodium), PyTorch, scikit-learn, PySpark, NumPy, Pandas, Feature Engineering, Agile/Scrum

EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer Intern

- Built enterprise agentic AI application (LangGraph, FastAPI, Python) - full RAG pipeline with document ingestion, embeddings, vector search, and reranking; MCP-compatible tool interfaces; deterministic guardrails; CI/CD-integrated eval gate with observability on retrieval quality and latency.
- Implemented multi-agent orchestration with HITL escalation; tracing and monitoring across agent performance and system reliability.

George Mason University

Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built Python ML pipelines (PyTorch, scikit-learn) on WHO and UK Road Safety datasets - $R^2 = 0.977$; end-to-end from data ingestion through feature engineering and statistical validation across 190+ countries.
- Mentored 600+ students in Python, TypeScript, SQL, and software engineering fundamentals; designed monitoring dashboards improving research comprehension by 40%.

Entrata - PropTech / Affordable Housing SaaS

Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **Full-stack production systems and APIs at enterprise scale** - 20+ scalable Python and JavaScript/TypeScript modules; FastAPI REST APIs and microservice integration layers; 99.5% uptime SLA across 100K+ monthly active users; Docker containerization (50% scalability improvement); AWS cloud-native infrastructure (S3, EC2, RDS, Glue).
- **AI-powered observability and data pipeline engineering** - AI-driven monitoring and tracing tooling reducing triage time 40%; owned HUD data pipeline 7 years (federal source → 180,000 domain records); RabbitMQ event-driven async processing with retry, idempotency, and dead-letter guarantees; Redis caching; 200+ SQL query optimizations (30% performance gain).
- **CI/CD, security, and operational excellence** - CI/CD via GitHub Actions with automated quality gates; 100% PHI compliance (Libsodium-encrypted pipelines, zero security incidents); 300+ code reviews; mentored 5+ engineers on production Python, distributed systems, and API design best practices.

Entrata India - Pune

Jan 2017 - Sep 2018

Associate Software Engineer → Trainee

- Built backend modules (PHP, Python, PostgreSQL, NoSQL); data pipeline for 1.6M HUD/Tax Credit records; AWS S3 migration resolving 90% of legacy storage constraints.

PROJECTS

AgentProbe: LLM Evaluation Framework with Observability & Tracing

Evals · Tracing · CI/CD

Python · FastAPI · LangGraph · ChromaDB · CI/CD (GitHub Actions) · Anthropic · OpenAI · Ollama | github.com/venkatalakshmiparimala/AgentProbe

Production LLM evaluation and observability platform: 15-endpoint FastAPI service, 5-metric quality engine (correctness, groundedness, latency, token efficiency, hallucination rate), LLM-as-judge scoring, CI-integrated regression gate, multi-provider routing. Full tracing and monitoring across AI application performance - zero regressions shipped across 1,240+ test cases. Open-source.

Leasel: 8-Agent RAG Platform with MCP & Agentic Workflows

RAG · Agentic · MCP

Python · LangGraph (8-agent) · FastAPI · TF-IDF · ChromaDB · OCR · SQLite | github.com/venkatalakshmiparimala/Lease_complaince_rag

Enterprise agentic AI platform: full RAG pipeline (document ingestion, OCR, indexing, embeddings, vector search, reranking, retrieval optimization, grounding) with MCP-compatible tool interfaces; 8-agent LangGraph stateful workflow with routing, planning, tool use, and state management; deterministic guardrails; HITL escalation; source-linked answers with 100% citation correctness. 93.6% precision, 87.9% F1 across 1,240 test cases.

AI on FHIR: Full-Stack NLP Enterprise Query System

Full-Stack · NLP · APIs

Python · Flask · React.js · TypeScript · Docker · BERT · spaCy · LLMs · FHIR APIs | github.com/venkatalakshmiparimala/AI_on_FHIR

Full-stack enterprise AI application (Python backend + React.js/TypeScript frontend) translating natural-language queries into FHIR API calls via BERT embeddings and LLM inference - semantic search reducing retrieval time ~60%; HIPAA-compliant RBAC with authentication, authorization, and full data isolation.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017